Data Golf 2026: how did 13 teams answer the same 18 questions?

Purpose. Compare the final submissions from the Data Golf 2026 Kaggle competition, find where teams agreed and where they diverged, and pick a few findings for a lightning talk at a SNOMED CT conference.

The competition in one paragraph. Teams answered 18 yes/no clinical questions for each of 279,912 synthetic patient encounters coded in SNOMED CT (AU editions 2021 to 2026). They used FHIR terminology services and ECL. The official score was the unweighted mean of per-question F1 (macro F1). Ground truth comes from the competition's `solution.csv`.

How to read this notebook. Each step has a short note on what was done and why, then the code, then (after the code) what it showed. Charts appear inline and are saved to `outputs/` as separate images for slides.

Conventions

- Correctness is defined only by the competition's solution file. SNOMED CT's own modelling is not treated as ground truth. Where a split looks like a genuine modelling ambiguity, we say so rather than pick a side.
- Fifteen submissions were received. The two lowest-ranked (Teams 14 and 15) are partial submissions: they appear in the validation and leaderboard tables and are left out of every later step. Set `EXCLUDE_BOTTOM = 0` to keep them.
- Teams are labelled **Team 1 to Team 15** by recomputed rank. No chart or table names a team. The last cell reveals the podium. The Kaggle rules promised that nothing identifying participants would appear in the Expo presentation, so please keep it that way for the rest.
- Placings are **recomputed here** from the submissions (unweighted macro F1 over all rows). They are not the Kaggle leaderboard. If the official order differs, set `TEAM_ORDER` in the setup cell.
- The organisers' expected difficulty (1 easiest to 6 hardest) comes from `Question_Rationale.md`. These are pre-event expectations, not measured results.
- Spelling follows Australian English.

Requirements. Python 3.10+, `numpy`, `pandas`, `matplotlib`, `scipy`. Run from the `post-event analysis` folder. The notebook reads `submissions/*.csv`, `Question_Rationale.md`, `../solution.csv`, `../Questions.md` and `../TrainingDataSet.csv`.

0. Setup

Question	Criterion
Q1	Patients with female as their recorded sex.
Q2	Patients with a BMI of 30 or more and a total cholesterol to HDL r...
Q3	Patients documented as being healthy or free of current problems.
Q4	Patients with hyperthyroidism according to their clinical note.
Q5	Patients with osteoarthritis in 2024.
Q6	Patients taking an ACE inhibitor.
Q7	Patients taking a product containing a cardiac (digitalis) glycoside.
Q8	Patients with a fungal foot infection.
Q9	Patients taking more than one product that contains paracetamol.
Q10	Patients with a bilateral condition.
Q11	Female patients aged 15 to 49 taking methotrexate.
Q12	Pregnant patients prescribed a teratogenic medicine.
Q13	Patients with osteoarthritis or rheumatoid arthritis taking an ora...
Q14	Patients with an open wound on an upper limb, excluding the finger...
Q15	Pregnant patients with a fasting glucose of 5.1 mmol/L or greater.
Q16	Patients with a non-Streptococcal bacterial joint infection.
Q17	Patients taking an oral iron supplement.
Q18	Patients taking mineral supplement excluding the electrolytes sodi...

1. Load and validate

What and why. Every submission is read into one boolean array shaped `[team, question, patient]`. Before anything else we check that each file covers the same patients as the solution file, uses clean True/False values, and has no missing cells. Problems are reported, never silently dropped.

Known quirks handled by the loader: values arrive as `TRUE`, `True` or `true`; one file starts with a byte-order mark; one has a quoted header. Teams are then ranked by recomputed macro F1 and relabelled, so no later output needs file names.

```
15 submissions x 18 questions x 279,912 patients. Public rows: 139,956; private rows: 139,956.
```

	rows	ids_match	same_order	duplicates	blank_cells	bom	spellings
Team 1	279912	True	True	0	0	False	false, true
Team 2	279912	True	True	0	0	False	False, True
Team 3	279912	True	True	0	0	False	False, True
Team 4	279912	True	True	0	0	False	False, True
Team 5	279912	True	True	0	0	False	False, True
Team 6	279912	True	True	0	0	False	False, True
Team 7	279912	True	True	0	0	False	False, True
Team 8	279912	True	True	0	0	False	False, True
Team 9	279912	True	True	0	0	False	False, True
Team 10	279912	True	True	0	0	False	False, True
Team 11	279912	True	True	0	0	False	False, True
Team 12	279912	True	True	0	0	False	False, True
Team 13	279912	True	True	0	0	True	FALSE, TRUE
Team 14	279912	True	True	0	0	False	False, True
Team 15	279912	True	True	0	0	False	FALSE, TRUE

Validation result. The table above is the evidence: row counts, ID matches, duplicates, blank cells and the spellings of True/False found in each file. The loader stops on any ID mismatch rather than dropping rows. The cell below turns that into one explicit statement.

Finding. All 15 files cover the same 279,912 Episode Numbers as the solution file, in the same order, with no duplicates or blanks. Nothing was dropped or realigned. The loader did have to handle three spellings of True/False and one file with a byte-order mark. Confidence: high.

```
All 15 submissions cover the same 279,912 Episode Numbers as the solution file, with no duplicates or blanks.
Files whose rows were in a different order (aligned by ID): 0
```

Recomputed leaderboard

What and why. We score every team with the competition's metric (unweighted mean of per-question F1) over all rows, and also over the public and private halves. The ranking is used to order teams in every later chart. It is **recomputed**, so confirm it against the official leaderboard before presenting it as final.

```
Leaderboard shows all 15 teams. Analysis from here on uses the top 13.
```

Rank	Team	Macro F1 (all rows)	Public	Private	Weakest question F1	Questions with F1 >= 0.99	Questions answered all False	Share of answers True (all cells)	In analysis
1	Team 1	0.9922	0.9918	0.9926	0.9426	14	0	0.0575	yes
2	Team 2	0.9914	0.9919	0.9909	0.9503	14	0	0.0574	yes
3	Team 3	0.9898	0.9906	0.9890	0.8972	14	0	0.0573	yes
4	Team 4	0.9685	0.9681	0.9689	0.8110	11	0	0.0577	yes
5	Team 5	0.9626	0.9620	0.9633	0.7219	12	0	0.0578	yes
6	Team 6	0.9497	0.9502	0.9493	0.5836	12	0	0.0574	yes
7	Team 7	0.9455	0.9455	0.9455	0.6510	10	0	0.0586	yes
8	Team 8	0.9328	0.9351	0.9306	0.6271	9	0	0.0562	yes
9	Team 9	0.9243	0.9266	0.9218	0.3956	9	0	0.0570	yes
10	Team 10	0.9106	0.9114	0.9098	0.5355	10	0	0.0561	yes
11	Team 11	0.8670	0.8680	0.8660	0.0000	10	0	0.0568	yes
12	Team 12	0.8623	0.8658	0.8586	0.3563	5	0	0.0553	yes
13	Team 13	0.8600	0.8623	0.8577	0.5061	8	0	0.0566	yes
14	Team 14	0.5657	0.5658	0.5656	0.0000	6	7	0.0457	no (partial)
15	Team 15	0.2168	0.2167	0.2168	0.0000	2	0	0.4019	no (partial)

Finding.

- The top three teams finish within 0.0024 of each other (macro F1 0.9922, 0.9914 and 0.9898). Teams 1 and 2 swap places between the public and private halves, so treat them as tied.
- Teams 14 and 15 look like partial submissions rather than weaker methods. Team 14 answered seven of the 18 questions as all False. Team 15 answered True on about 40% of all cells, against about 6% for every other team, which looks like random fill. The last column of the table marks both "no (partial)".
- They are left out of every step after this one. With them in, mean percent agreement between teams was about 0.93 on most questions. With them out it is above 0.98 on every question (lowest 0.988, on Q18). They also create apparent majorities that only reflect their columns. The validation and leaderboard tables keep all 15 so the reason stays visible, and step 2 shows the before and after side by side.
- Confidence: high for the table (computed from the files). The ranking is recomputed, not the official leaderboard. Calling Teams 14 and 15 "partial" is our reading of the patterns above; the organisers' records may say more.

2. Question-level agreement

What and why. For every question we compare all 78 pairs of the 13 analysed teams using two measures:

- **Percent agreement**: the share of patients on which two teams gave the same answer.
- **Cohen's kappa**: agreement beyond what chance would produce.

We use both because most questions have a rare "True" class. For Q7 only 239 of 279,912 patients are True, so two teams that disagree on every True patient still agree on over 99.9% of rows. Percent agreement alone would hide that. Kappa is the primary ranking measure for that reason.

We also compute each team's F1 against ground truth, and rank the questions from most to least contested (lowest mean pairwise kappa first).

Question	Criterion	expected_difficulty	truth_true	mean_agreement	mean_kappa	mean_F1	F1_sd	F1_min	F1_max	te
Q12	Pregnant + teratogenic medicine	6	347	0.9986	0.4565	0.6273	0.2724	0.0000	0.9913	
Q15	Pregnant + fasting glucose >= 5.1	4	461	0.9987	0.5811	0.6813	0.2332	0.0172	0.9538	
Q18	Mineral supplement (not Na/K/Cl)	5	14498	0.9876	0.8428	0.8965	0.0970	0.7427	0.9946	
Q16	Non-Streptococcal bacterial joint infection	3	1472	0.9984	0.8906	0.9393	0.1255	0.5665	1.0000	
Q13	OA/RA + oral NSAID + eGFR < 60	4	1819	0.9990	0.8995	0.9069	0.1014	0.5825	0.9646	
Q17	Oral iron supplement	3	10649	0.9947	0.9134	0.9549	0.0814	0.7634	1.0000	
Q4	Hyperthyroidism (clinical note)	4	10859	0.9932	0.9253	0.9363	0.0686	0.7769	1.0000	
Q5	Osteoarthritis in 2024	3	3819	0.9985	0.9382	0.9451	0.0692	0.7099	0.9995	
Q9	More than one paracetamol product	4	3230	0.9993	0.9621	0.9812	0.0642	0.7589	1.0000	
Q14	Open wound, upper limb (not fingers/thumb)	3	22600	0.9961	0.9733	0.9862	0.0154	0.9571	1.0000	
Q3	Healthy / no current problems	2	15731	0.9984	0.9832	0.9920	0.0276	0.8963	1.0000	
Q11	Female 15-49 on methotrexate	4	2250	0.9998	0.9869	0.9935	0.0196	0.9264	1.0000	
Q6	ACE inhibitor	3	9059	0.9995	0.9922	0.9962	0.0131	0.9508	1.0000	
Q8	Fungal foot infection	3	179	1.0000	0.9933	0.9966	0.0117	0.9563	1.0000	
Q10	Bilateral condition	4	7038	0.9999	0.9984	0.9991	0.0013	0.9951	1.0000	
Q2	BMI 30+ and chol/HDL > 5	2	45542	1.0000	0.9999	1.0000	0.0001	0.9997	1.0000	
Q1	Female	1	139606	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	
Q7	Cardiac (digitalis) glycoside	3	239	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	

Finding.

- Q1, Q2 and Q7 are not contested. Q1 and Q7 are perfect for every team, and Q2 scores F1 0.9997 or better. Q7 only looked contested when the partial submissions were included.
- Q12 and Q15, the two pregnancy questions, are the most contested by a wide margin (mean kappa 0.46 and 0.58). Mean F1 is only 0.63 and 0.68. No team reached F1 0.99 on Q15, and only one did on Q12.
- Q18 is third (kappa 0.84, mean F1 0.90), followed by Q16 and Q13 (kappa 0.89 and 0.90). Below Q5, every question has mean kappa above 0.96.
- Using only the top 10 teams moves ranks by at most five places (column `rank_shift`). The top three do not move.
- Confidence: high for Q12, Q15 and Q18. Medium below that, where kappa differences are small (0.92 to 0.99) and ranks move by up to five places depending on which teams are included.

What did excluding the two partial submissions change?

What and why. To show whether dropping Teams 14 and 15 was justified, we repeat the question-level comparison with all 15 teams and set it beside the 13-team version. For each question we compare mean percent agreement, mean kappa, mean F1 and the "most contested" rank (1 = most contested).

	Criterion	agreement_15	agreement_13	kappa_15	kappa_13	mean_F1_15	mean_F1_13	contested_rank_15	contes
Q12	Pregnant + teratogenic medicine	0.9323	0.9986	0.3391	0.4565	0.5438	0.6273	1	
Q15	Pregnant + fasting glucose >= 5.1	0.9322	0.9987	0.4316	0.5811	0.5906	0.6813	2	
Q18	Mineral supplement (not Na/K/Cl)	0.9187	0.9876	0.6261	0.8428	0.7833	0.8965	3	
Q16	Non-Streptococcal bacterial joint infection	0.9314	0.9984	0.6616	0.8906	0.8147	0.9393	4	
Q13	OA/RA + oral NSAID + eGFR < 60	0.9319	0.9990	0.6682	0.8995	0.7869	0.9069	5	
Q17	Oral iron supplement	0.9278	0.9947	0.7776	0.9134	0.8847	0.9549	8	
Q4	Hyperthyroidism (clinical note)	0.9917	0.9932	0.9073	0.9253	0.9261	0.9363	16	
Q5	Osteoarthritis in 2024	0.9322	0.9985	0.8166	0.9382	0.8846	0.9451	9	
Q9	More than one paracetamol product	0.9327	0.9993	0.8362	0.9621	0.9186	0.9812	10	
Q14	Open wound, upper limb (not fingers/thumb)	0.9204	0.9961	0.7231	0.9733	0.8640	0.9862	6	
Q3	Healthy / no current problems	0.9909	0.9984	0.8855	0.9832	0.9422	0.9920	15	
Q11	Female 15-49 on methotrexate	0.9331	0.9998	0.8560	0.9869	0.9287	0.9935	12	
Q6	ACE inhibitor	0.9329	0.9995	0.8604	0.9922	0.9341	0.9962	13	
Q8	Fungal foot infection	0.9999	1.0000	0.8469	0.9933	0.9223	0.9966	11	
Q10	Bilateral condition	0.9301	0.9999	0.7417	0.9984	0.8691	0.9991	7	
Q2	BMI 30+ and chol/HDL > 5	1.0000	1.0000	0.9999	0.9999	1.0000	1.0000	17	
Q1	Female	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	18	
Q7	Cardiac (digitalis) glycoside	0.9334	1.0000	0.8667	1.0000	0.9334	1.0000	14	

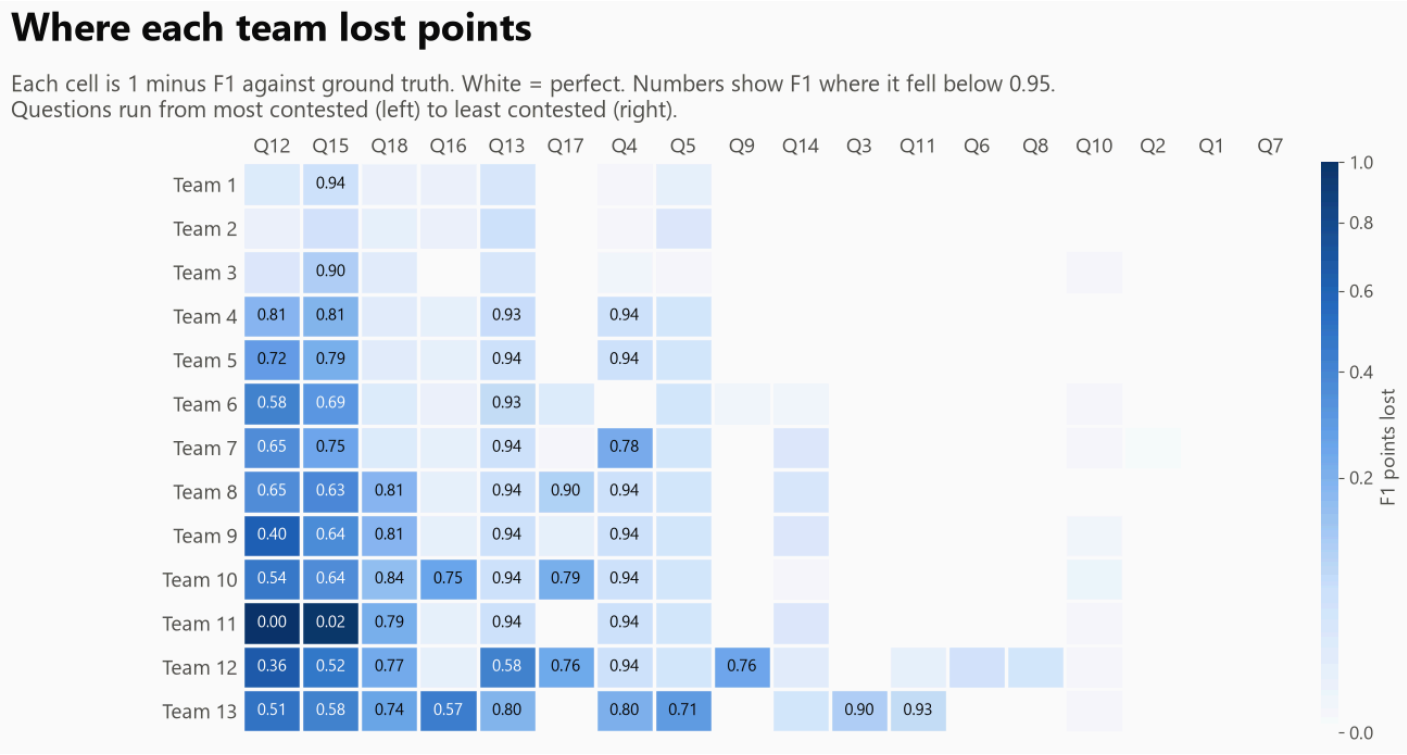
```
Mean percent agreement between teams: median across questions 0.933 with all 15, 0.999 with 13.
Questions with mean agreement below 0.95: 13 with all 15, 0 with 13.
Questions where mean F1 changes by more than 0.05 when the two are removed: Q12, Q15, Q18, Q16, Q13, Q17, Q5, Q9, Q14, Q11,
Q6, Q8, Q10, Q7
Questions whose contested rank moves by 4 or more places: Q4 (16 -> 7); Q14 (6 -> 10); Q3 (15 -> 11); Q10 (7 -> 15)
```

Finding. Keeping Teams 14 and 15 gave several misleading impressions, which is why they are left out. The five most contested questions (Q12, Q15, Q18, Q16 and Q13) are the same, in the same order, either way, so the headline result does not depend on this choice. The details below do.

- **Disagreement was overstated.** The median of mean percent agreement across questions was 0.933 with all 15 teams and 0.999 with 13. Thirteen questions had mean agreement below 0.95 with all 15 teams, and none do with 13. Most of that apparent disagreement came from two columns, not from the other teams.
- **Some questions looked contested when they were not.** Q10 ranked 7th most contested with 15 teams and 15th with 13: every one of the 13 complete teams scored F1 0.995 or better. Q7 (mean kappa 0.87 to 1.00) and Q14 (6th to 10th) also changed.
- **One looked uncontested when it was not.** Q4 ranked 16th with 15 teams and 7th with 13. Its kappa barely moved (0.91 to 0.93) while the other questions' kappas rose a great deal, so the real disagreement among the complete teams (see Finding 1) had been hidden.
- **Difficulty was overstated.** Mean F1 changes by more than 0.05 on 14 of 18 questions. Q10 goes from 0.87 to 0.999 and Q14 from 0.86 to 0.99. This matters for step 3, because difficulty read off all 15 teams would look much higher than it was for teams that submitted complete answers.
- **Confidence: high** that the two submissions distort the comparison, since every number above is computed from the files. Calling them "partial" is still our judgement. Set `EXCLUDE_BOTTOM = 0` to see the notebook with them included.

Chart A: where each team lost points

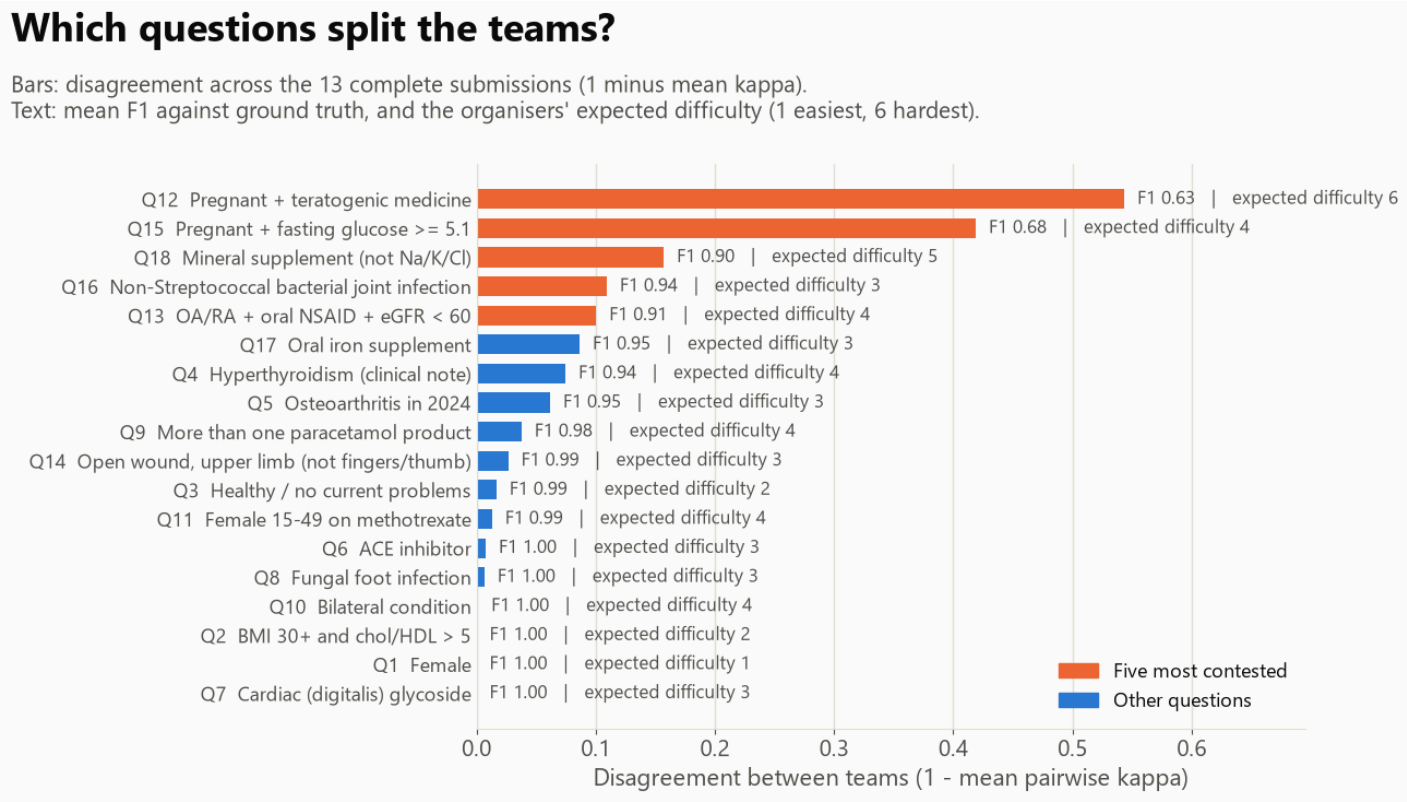
What and why. One cell per team and question showing how many F1 points the team lost against ground truth (1 minus F1). White means a perfect answer, so the eye goes straight to the failures. Columns run from the most contested question (left) to the least. Cells below F1 0.95 show their F1.



Reading Chart A. Teams 14 and 15 are not shown (partial submissions). Teams 1 to 3 are almost white across the board; their weakest cell is Team 3 on Q15 (F1 0.90). Teams 4 to 10 lose most of their points on the two pregnancy questions (F1 from 0.40 to 0.81), and most share a small loss on Q4 and Q13 (F1 about 0.94 each). Team 11 scored almost nothing on Q12 and Q15 (F1 0.00 and 0.02). Team 13 loses points on the most questions (nine cells below F1 0.95). Confidence: high.

Chart B: the most contested questions

What and why. Questions ranked by how much teams disagreed with each other (1 minus mean pairwise kappa). The five most contested are highlighted. The text at the end of each bar gives the mean F1 against ground truth, so you can see whether disagreement also meant being wrong. Each bar also shows the organisers' expected difficulty from `Question_Rationale.md`.



Reading Chart B. Q12, Q15 and Q18 stand clear of the rest. The two highest expected ratings, Q12 (6) and Q18 (5), are both among the three most contested. Q15 was rated 4. The next five bars (Q16, Q13, Q17, Q4 and Q5) are much shorter, and questions rated 3 or 4 appear all the way down the chart. Step 3 tests this properly.

3. Did the questions turn out as hard as the organisers expected?

What and why. Before the event the organisers rated each question from 1 (easiest) to 6 (hardest) in `Question_Rationale.md`. Here we compare those ratings with what happened. We use four measures of observed difficulty, all oriented so that a higher number means harder:

- **F1 lost**: 1 minus the mean F1 across teams.
- **Disagreement**: 1 minus the mean pairwise kappa between teams.
- **Spread**: the standard deviation of team F1.
- **Share of teams below F1 0.95**.

We report **Spearman** and **Kendall** rank correlations, because the ratings are ordinal and many questions share a rating (seven questions are rated 3, six are rated 4). A second set leaves out Q1 and Q2, the two questions that need no terminology work, to check that the easy end is not carrying the result. A third checks how far one question can move the answer: drop each question in turn and look at the range of Spearman values.

Then we rank questions by expected and observed difficulty and flag any that differ by 6 places or more.

Observed difficulty measure	Spearman rho (18 questions)	p	Kendall tau-b	p (tau)	Spearman rho (without Q1, Q2)	p (without Q1, Q2)	rho range, drop one question
F1 lost (1 - mean F1)	0.667	0.002	0.565	0.003	0.548	0.028	0.60 to 0.76
Disagreement (1 - mean kappa)	0.625	0.006	0.520	0.006	0.487	0.056	0.55 to 0.71
Spread (SD of team F1)	0.586	0.011	0.474	0.012	0.431	0.095	0.50 to 0.67
Share of teams below F1 0.95	0.631	0.005	0.523	0.009	0.560	0.024	0.57 to 0.72

```
Using only the top 10 teams: Spearman rho between expected difficulty and F1 lost = 0.65 (p = 0.003).

Without the 1 hardest by outcome (Q12): Spearman rho = 0.60 (p = 0.011)
Without the 2 hardest by outcome (Q12, Q15): Spearman rho = 0.57 (p = 0.022)
Without the 3 hardest by outcome (Q12, Q15, Q18): Spearman rho = 0.47 (p = 0.079)
```

	criterion	skill (organiser)	expected_difficulty	F1_lost	expected_rank	observed_rank
Q16	Non-Streptococcal bacterial joint infection	Attributes combined with subtraction	3	0.061	12.0	6.0
Q3	Healthy / no current problems	First ECL; a concept outside the obvious hierarchy	2	0.008	16.5	11.0
Q5	Osteoarthritis in 2024	Large edition drift	3	0.055	12.0	7.0
Q17	Oral iron supplement	Substance grouper and dose form	3	0.045	12.0	8.0
Q15	Pregnant + fasting glucose >= 5.1	Pregnancy identification plus a lab value	4	0.319	5.5	2.0
Q14	Open wound, upper limb (not fingers/thumb)	ECL using the anatomy hierarchy	3	0.014	12.0	10.0
Q13	OA/RA + oral NSAID + eGFR < 60	Drift, complex ECL and dose form	4	0.093	5.5	4.0
Q1	Female	Basic lookup	1	0.000	18.0	17.5
Q2	BMI 30+ and chol/HDL > 5	Data wrangling, no SNOMED CT	2	0.000	16.5	16.0
Q4	Hyperthyroidism (clinical note)	Edition drift; a 2022 hierarchy change; precision	4	0.064	5.5	5.0
Q12	Pregnant + teratogenic medicine	Pregnancy identification and a self-built drug list	6	0.373	1.0	1.0
Q6	ACE inhibitor	Drug class inferred through ingredients	3	0.004	12.0	13.0
Q18	Mineral supplement (not Na/K/Cl)	Building your own enumerated list	5	0.103	2.0	3.0
Q8	Fungal foot infection	Composing an ECL where no single concept exists	3	0.003	12.0	14.0
Q9	More than one paracetamol product	Counting across medicine columns	4	0.019	5.5	9.0
Q7	Cardiac (digitalis) glycoside	Drug class inferred through ingredients	3	0.000	12.0	17.5
Q11	Female 15-49 on methotrexate	Multiple criteria with calculated age	4	0.006	5.5	12.0
Q10	Bilateral condition	Complex ECL with left and right finding sites	4	0.001	5.5	15.0

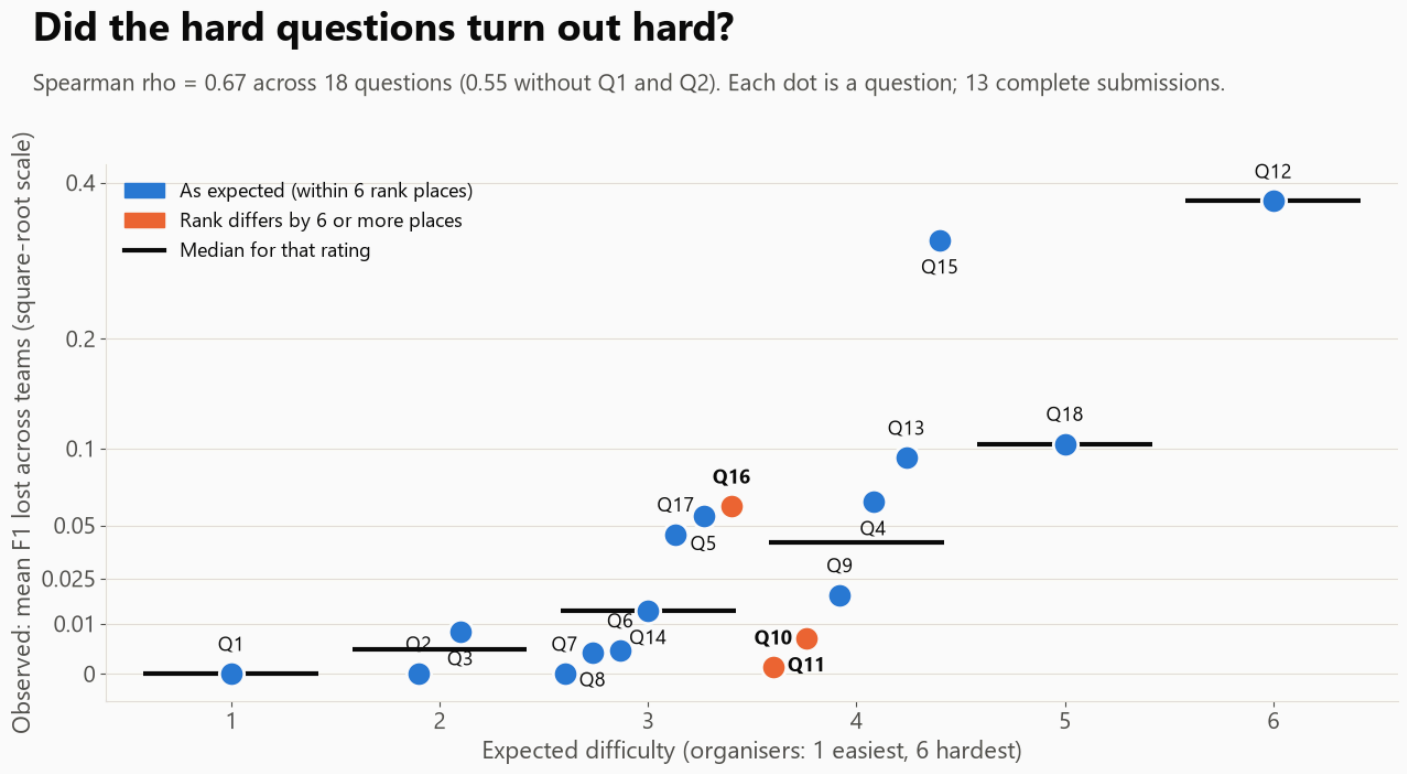
Finding.

- **Broadly, yes.** The organisers' ratings track what happened. Against F1 lost, Spearman rho is 0.67 (p = 0.002) and Kendall tau-b is 0.57 (p = 0.003). The other three measures give rho between 0.59 and 0.63, all with p below 0.02.
- **The result is not carried by one question.** Dropping any single question leaves rho between 0.60 and 0.76. Using only the top 10 teams gives 0.65. Without Q12 it is 0.61, and without Q12 and Q15 it is 0.57 (p = 0.02).
- **It does lean on the hardest end.** Without the three hardest questions by outcome (Q12, Q15 and Q18), rho falls to 0.47 and is no longer significant at 5% (p = 0.08). Without Q1 and Q2, the two questions that need no terminology work, rho is 0.55 for F1 lost (p = 0.03) but 0.49 for disagreement (p = 0.06) and 0.43 for spread (p = 0.10). The ratings are good at picking out the hard questions and weaker at ordering the middle.
- **The top of the scale matches.** Q12 was rated 6 and was the hardest by outcome. Q18 was rated 5 and was third. Q15 was rated 4 and was second hardest.
- **Questions that differ by 6 or more rank places:**
  - Q16 (rated 3, sixth hardest by outcome) was harder than expected.
  - Q11 and Q10 (both rated 4, twelfth and fifteenth) were easier than expected. Q10, rated for its "complex ECL with left and right finding sites", lost almost nothing: every one of the 13 teams scored F1 0.995 or better.
  - Q3 (harder than expected by 5.5 places) and Q7 (easier by 5.5) just miss the cut.
- **Confidence: medium.** There are 18 questions on a six-point scale with many ties, so a single rank place is not meaningful, and a gap of 6 places is a rule of thumb rather than a test. Difficulty was measured on

the 13 stronger teams, which may compress the middle of the scale. We have not tested that.

Chart E: expected difficulty against what happened

What and why. Each dot is one question. Left to right is the organisers' expected difficulty; height is how many F1 points the teams lost on average, on a square-root scale so that the crowded low end can be read. If the ratings predicted difficulty, the dots would climb from left to right. The black bar marks the median of each rating. Orange dots are the questions flagged above as differing by 6 or more rank places.



Reading Chart E. The dots rise from left to right, but unevenly. Ratings 1 to 3 sit near zero. The questions rated 4 range from almost no loss (Q10, Q11) to the second largest (Q15). The two outliers, Q12 and Q15, both involve identifying pregnant patients, and `Question_Rationale.md` calls that the hardest part of each. Q15 was rated 4 but turned out to be the second hardest question by a wide margin. The 6-place rule does not flag it, because six questions share rating 4 and their tied rank (5.5) is already near the top.

4. Team clustering

What and why. We measure how similar two teams are across all 18 questions at once, then cluster them. The distance is **1 minus the mean per-question kappa**, not plain percent agreement over all cells. Pooled percent agreement would be above 98% for almost every pair because most cells are easy True-negatives, and every team would look alike. As a check we repeat the clustering with plain percent agreement and report how much the two trees agree.

We then ask whether the top-scoring teams sit together (a convergent strategy) or spread across clusters (several valid approaches). With only 13 teams this is **low confidence**.

```
Cophenetic correlation between the kappa-based and percent-agreement-based trees: 0.83
```

Team	Rank	Macro F1	Cluster (4)
Team 1	1	0.9922	1
Team 2	2	0.9914	1
Team 3	3	0.9898	1
Team 4	4	0.9685	1
Team 5	5	0.9626	1
Team 6	6	0.9497	1
Team 7	7	0.9455	1
Team 8	8	0.9328	1
Team 9	9	0.9243	1
Team 10	10	0.9106	1
Team 11	11	0.8670	2
Team 12	12	0.8623	3
Team 13	13	0.8600	4

```
Mean distance within the top 5 teams: 0.026
Mean distance between top 5 and the other 10: 0.090
Mean distance within the other 10: 0.119
```

Finding.

- The top five teams are close to one another (mean distance 0.026). The other eight are further from the top five (0.090) and from each other (0.119). The top ten teams form one cluster, and Teams 11, 12 and 13 each sit in a cluster of their own. The top teams are a **convergent group**, not several different approaches.
- The kappa-based tree and the percent-agreement tree agree reasonably well (cophenetic correlation 0.83), so the grouping does not depend heavily on the distance chosen.
- Confidence: medium. Thirteen teams is a small sample and we have not seen the teams' methods, so "similar answers" is not proof of "similar approach".

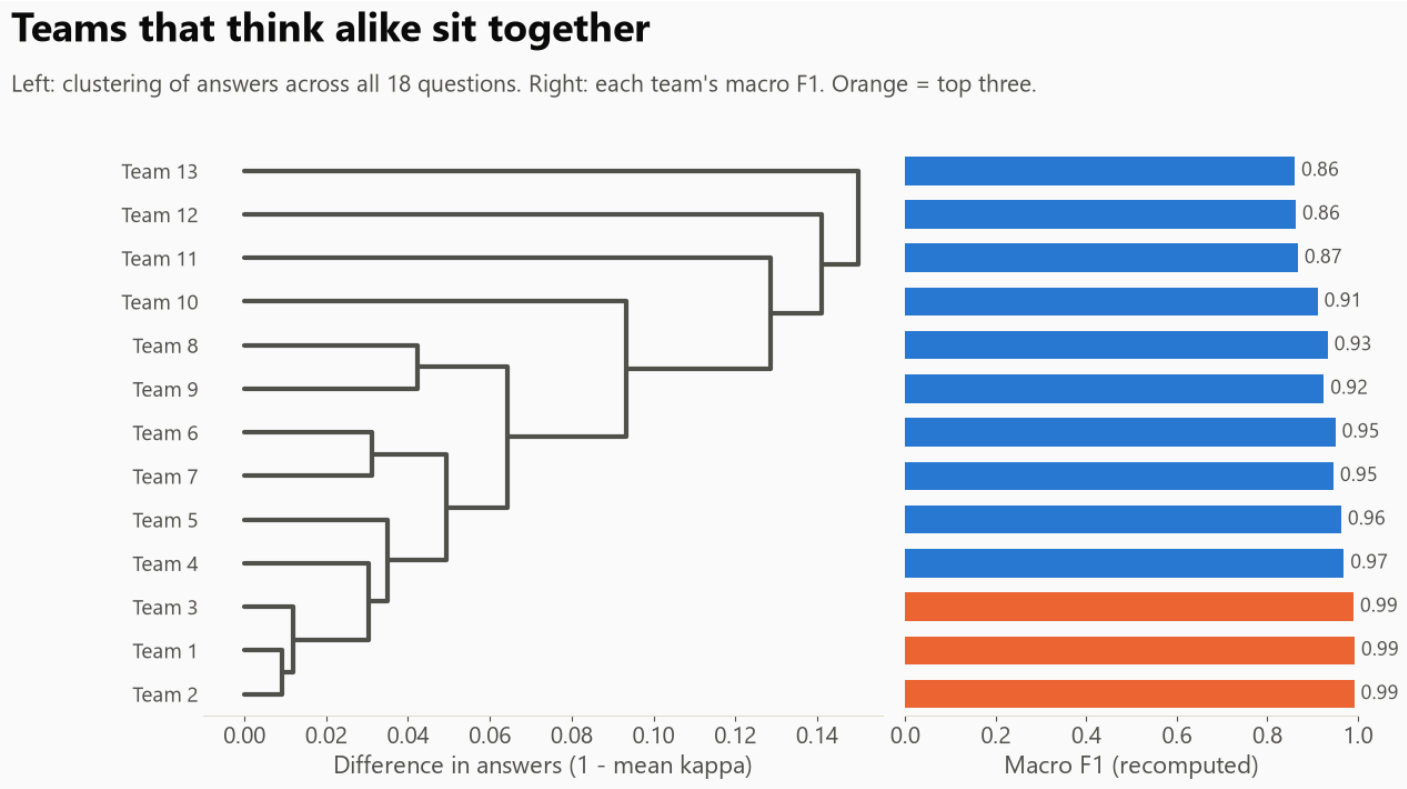
Answer twins

What and why. Kappa shows that teams are similar, but not whether they gave *exactly* the same answers. For each question we group teams whose answer columns are identical, row for row. A large group of identical columns that are not perfect means the same mistake was made independently, which is a strong sign of a shared approach (for example, the same ECL).

Question	Criterion	distinct_answer_columns	largest_identical_group	largest_identical_group_not_perfect	F1_of_that_grou
Q5	Osteoarthritis in 2024	6	8	8	0.957
Q4	Hyperthyroidism (clinical note)	6	7	7	0.943
Q16	Non-Streptococcal bacterial joint infection	9	5	5	0.988
Q13	OA/RA + oral NSAID + eGFR < 60	9	5	5	0.944
Q14	Open wound, upper limb (not fingers/thumb)	8	5	2	0.970
Q18	Mineral supplement (not Na/K/Cl)	12	2	2	0.814
Q2	BMI 30+ and chol/HDL > 5	3	11	0	Na
Q1	Female	1	13	0	Na

Finding. On Q5, eight teams (4, 5, 7, 8, 9, 10, 11 and 12) gave exactly the same answer on all 279,912 rows, and all scored F1 0.9574. On Q4, seven teams (4, 5, 8, 9, 10, 11 and 12) did the same, all scoring 0.9433. Identical wrong columns almost certainly mean the same rule or the same query. Confidence: high that the answers are identical; medium that they share a method.

Chart C: how similar were the teams' answers?



5. Row-level contested cases

What and why. For every patient and question we count how many of the 13 analysed teams said True. That separates two kinds of contested row:

- **Shared misconception**: at least two thirds of the teams (9 of 13) gave the same answer and it was wrong.
- **Genuine split**: between one third and two thirds of the teams (5 to 8 of 13) said True and the row is not a shared misconception. These are boundary cases where the teams could not agree.

Who votes. All 13 analysed teams vote. Teams 14 and 15 are excluded because their columns are all False or close to random, which would create apparent "majorities" that only reflect those columns. As a robustness check the table below repeats the counts with the top 10 teams only.

To make this useful for a SNOMED CT audience, rows are grouped by the **concepts** they carry (the clinical note concept and the medicine concepts) rather than shown one patient at a time. For each concept the tables show how many flagged rows carry it, what share of that concept's rows were flagged, and how often ground truth says True among them. Concepts are ranked by `driver_score` = flagged rows x share flagged, which favours concepts that account for many contested rows *and* are consistently contested.

Confidence: HIGH = 200 or more rows, MEDIUM = 30 to 199, LOW = under 30. Remember that a concept can appear in a contested row only because it sits beside the concept that really causes the problem (for example a clinical note beside a medicine). The `share_flagged` column helps to tell the two apart.

```
Voting teams: all 13 analysed teams.
Shared misconception: at least 9 of 13 teams give the same wrong answer.
Genuine split: between 5 and 8 of 13 teams say True, and not a misconception.
```

Question	Criterion	misconception_rows (all 13)	split_rows (all 13)	misconception_rows (top 10)	split_rows (top 10)
Q12	Pregnant + teratogenic medicine	7	198	3	198
Q15	Pregnant + fasting glucose >= 5.1	62	257	23	248
Q18	Mineral supplement (not Na/K/Cl)	337	3770	337	172
Q16	Non-Streptococcal bacterial joint infection	34	4	24	10
Q13	OA/RA + oral NSAID + eGFR < 60	172	32	198	3
Q17	Oral iron supplement	0	0	0	23
Q4	Hyperthyroidism (clinical note)	1301	0	30	1271
Q5	Osteoarthritis in 2024	340	0	340	0
Q9	More than one paracetamol product	0	0	0	0
Q14	Open wound, upper limb (not fingers/thumb)	0	55	0	0
Q3	Healthy / no current problems	0	0	0	0
Q11	Female 15-49 on methotrexate	0	0	0	0
Q6	ACE inhibitor	0	0	0	0
Q8	Fungal foot infection	0	0	0	0
Q10	Bilateral condition	0	10	0	10
Q2	BMI 30+ and chol/HDL > 5	0	0	0	0
Q1	Female	0	0	0	0
Q7	Cardiac (digitalis) glycoside	0	0	0	0

```
Saved the top 10 concepts per question and type to outputs/row_level_examples.csv (134 rows).
```

Shared misconceptions, most contested questions first

These are rows where at least two thirds of teams agreed with each other and were wrong.
`truth_true_share` is the share of those rows where ground truth is True.

```
Questions listed (most contested first). Misconceptions: ['Q15', 'Q18', 'Q16', 'Q13', 'Q4', 'Q5'] | splits: ['Q12', 'Q15',
'Q18', 'Q13', 'Q14']

Q15  Pregnant patients with a fasting glucose of 5.1 mmol/L or greater.
   shared-misconception rows: 62
```

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true	driver_
0	clinical note	Iron deficiency anaemia in mother complicating childbirth	25	182	0.137	1.0	4.0	
1	clinical note	Severe pre-eclampsia - delivered	8	43	0.186	1.0	4.0	
2	clinical note	Amniotic fluid cell content OK	2	3	0.667	1.0	3.0	
3	clinical note	Seven previous induced terminations of pregnancy	2	3	0.667	1.0	1.0	
4	clinical note	Helical ulceration of umbilical cord	1	1	1.000	1.0	3.0	

```
Q18  Patients taking mineral supplement excluding the electrolytes sodium, potassium or chloride.
   shared-misconception rows: 337
```

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_t
0	medicine	Miconazole nitrate 2.5 microgram/mg and zinc oxide 150 microgram/m...	98	98	1.0	1.0	
1	medicine	Daktozin ointment	45	45	1.0	1.0	
2	medicine	Resolve Nappy Rash ointment	39	39	1.0	1.0	
3	medicine	Amolin Baby cream (Old Formulation 2010)	29	29	1.0	1.0	
4	medicine	Soov Prickly Heat powder	27	27	1.0	1.0	

```
Q16  Patients with a non-Streptococcal bacterial joint infection.
   shared-misconception rows: 34
```

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_t
0	clinical note	Bacterial spondyloarthritis	10	10	1.000	1.0	
1	clinical note	Postinfective arthropathy following infection caused by Mycobacterium	8	10	0.800	1.0	
2	clinical note	Hydrarthrosis of yaws	5	5	1.000	1.0	
3	clinical note	Reactive arthropathy following infection caused by Mycobacterium	6	13	0.462	1.0	
4	clinical note	Infection of intervertebral disc caused by bacterium	2	2	1.000	1.0	

```
Q13  Patients with osteoarthritis or rheumatoid arthritis taking an oral NSAID, and their eGFR is below 60.
   shared-misconception rows: 172
```

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true
0	medicine	Ketorolac trometamol 5 mg/mL eye solution	25	928	0.027	1.0	0.159
1	clinical note	Calcification of cervical disc	3	18	0.167	0.0	11.000
2	clinical note	Chronic tophaceous gout of ankle and/or foot due to drug	2	13	0.154	0.0	9.000
3	clinical note	Chondrocalcinosis of right hand	2	13	0.154	0.0	10.000
4	clinical note	Chronic tophaceous gout of multiple sites due to renal impairment	2	14	0.143	0.0	9.000

Q4 Patients with hyperthyroidism according to their clinical note.
shared-misconception rows: 1,301

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true	driver_
0	clinical note	Familial dysalbuminaemic hyperthyroxinaemia	637	637	1.000	0.0	9.0	63
1	clinical note	Dysprealbuminaemic euthyroidal hyperthyroxinaemia	634	634	1.000	0.0	9.0	63
2	clinical note	Thyrotoxicosis due to pituitary thyroid hormone resistance	28	34	0.824	1.0	2.0	2
3	medicine	Thiamazole 15 mg oral tablet	86	988	0.087	0.0	9.0	
4	medicine	Carbimazole 5 mg oral tablet	79	1005	0.079	0.0	9.0	

Q5 Patients with osteoarthritis in 2024.
shared-misconception rows: 340

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true	driver_sco
0	clinical note	Bilateral gouty tophus of elbows	47	272	0.173	0.0	9.0	8.1
1	clinical note	Chondrocalcinosis of right hip	6	9	0.667	0.0	10.0	4.0
2	clinical note	Chronic tophaceous gout of knee due to drug	8	19	0.421	0.0	9.0	3.3
3	clinical note	Chondrocalcinosis due to dicalcium phosphate crystals	7	19	0.368	0.0	11.0	2.5
4	clinical note	Chronic tophaceous gout due to drug	6	16	0.375	0.0	9.0	2.2

Genuine splits, most contested questions first

These are rows where teams were roughly evenly divided.

Q12 Pregnant patients prescribed a teratogenic medicine.
split rows: 198

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_
0	clinical note	Transient hypertension of pregnancy - not delivered	33	45	0.733	0.303	5
1	clinical note	Toxaemia of pregnancy	9	15	0.600	0.000	5
2	clinical note	Pre-existing hypertension in obstetric context	5	6	0.833	0.600	6
3	clinical note	Severe pre-eclampsia - not delivered	12	38	0.316	1.000	6
4	clinical note	Chronic hypertension complicating AND/OR reason for care during pr...	7	13	0.538	0.000	6

Q15 Pregnant patients with a fasting glucose of 5.1 mmol/L or greater.
split rows: 257

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true	drive
0	clinical note	Iron deficiency anaemia of pregnancy	64	340	0.188	0.000	6.312	
1	clinical note	Maternal syphilis during pregnancy - baby not yet delivered	29	110	0.264	0.207	5.414	
2	clinical note	Transient hypertension of pregnancy - not delivered	16	45	0.356	0.312	5.938	
3	clinical note	Chronic hypertension in obstetric context	5	9	0.556	0.400	6.200	
4	clinical note	Hypertensive renal disease in obstetric context	3	4	0.750	0.667	7.000	

Q18 Patients taking mineral supplement excluding the electrolytes sodium, potassium or chloride.
split rows: 3,770

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_tr
0	medicine	Ferrous fumarate 305 mg and folic acid 350 microgram oral capsule	493	512	0.963	1.0	8
1	medicine	Ferric ammonium citrate 16 mg/mL and folic acid 100 microgram/mL o...	486	507	0.959	1.0	8
2	medicine	Ferrous sulfate 150 mg and folic acid 500 microgram prolonged-rele...	477	499	0.956	1.0	8
3	medicine	Ferric ammonium citrate 16 mg/mL and folic acid 500 microgram/mL o...	470	489	0.961	1.0	8
4	medicine	Ferrous fumarate 324 mg and folic acid 1 mg oral tablet	467	498	0.938	1.0	8

Q13 Patients with osteoarthritis or rheumatoid arthritis taking an oral NSAID, and their eGFR is below 60.
split rows: 32

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true	driver_score	conf
0	clinical note	Bilateral gouty tophus of elbows	13	272	0.048	0.0	8.0	0.621	
1	clinical note	Polyarticular acute gout caused by lead	2	11	0.182	0.0	8.0	0.364	
2	clinical note	Chronic tophaceous gout of wrist due to drug	2	16	0.125	0.0	8.0	0.250	
3	clinical note	Chronic tophaceous gout due to drug	2	16	0.125	0.0	8.0	0.250	
4	clinical note	Monoarticular acute gout caused by lead	1	14	0.071	0.0	8.0	0.071	

Q14 Patients with an open wound on an upper limb, excluding the fingers and thumb.
split rows: 55

	kind	display	rows_flagged	rows_total	share_flagged	truth_true_share	mean_teams_true	driver_score	cor
0	clinical note	Laceration of fascia of intrinsic muscle of finger	14	16	0.875	1.0	8.0	12.250	
1	clinical note	Laceration of intrinsic muscle of finger	14	19	0.737	1.0	8.0	10.316	
2	clinical note	Open fracture thumb metacarpal head	9	16	0.562	1.0	8.0	5.062	
3	clinical note	Open fracture thumb metacarpal neck	7	20	0.350	1.0	8.0	2.450	
4	clinical note	Open fracture of thumb metacarpal	4	13	0.308	1.0	8.0	1.231	

Finding. With all 13 analysed teams voting:

- *Shared misconceptions* (at least 9 of 13 wrong together) are concentrated in **Q4** (1,301 rows, led by the two hyperthyroxinaemia notes), **Q5** (340 rows, all in 2024, ground truth False, with gout and chondrocalcinosis-type notes among the most common), **Q18** (337 rows, led by skin products containing zinc, where only 2 of 13 teams on average said True and ground truth is True), **Q13** (172 rows spread over many concepts) and small numbers on Q15 (62) and Q16 (34).
- *Genuine splits* (5 to 8 of 13 teams saying True) are concentrated in **Q18** (3,770 rows, mostly iron-and-folic-acid products that five teams miss), **Q15** (257) and **Q12** (198), the pregnancy questions. Q14 has 55 and Q13 has 32.
- Q17, Q9, Q3, Q11, Q6, Q8, Q7, Q2 and Q1 have no flagged rows of either kind. Q10 has 10.
- **The voting pool matters for two questions.** On Q4, nine of 13 teams (69%) said True on the hyperthyroxinaemia notes, just over the two-thirds line, so the same rows are a "shared misconception" with 13 voters but a "genuine split" (1,271 rows) with the top 10. On Q18, split rows are 3,770 with 13 voters but 172 with the top 10, because most of the teams that miss the iron-and-folic-acid products are ranked 8 to 13.
- Confidence: high for the Q4 and Q18 counts (hundreds of rows per concept). Low for the Q15, Q16 and Q13 concept lists, where each concept has fewer than 30 flagged rows and may only sit next to the real cause. Those lists should not go on a slide.

6. Shortlist for the talk

What and why. The steps above point to four findings that a SNOMED CT audience will recognise from their own work. Each gets its own chart. For each one the evidence is computed below; the plain-language reading follows the chart.

Shared helpers: a strip chart that shows, for chosen groups of patients, how often each team's answer differed from ground truth. Groups are built from the clinical note and medicine names in `TrainingDataSet.csv`. Keyword groups are approximate and are labelled as such.

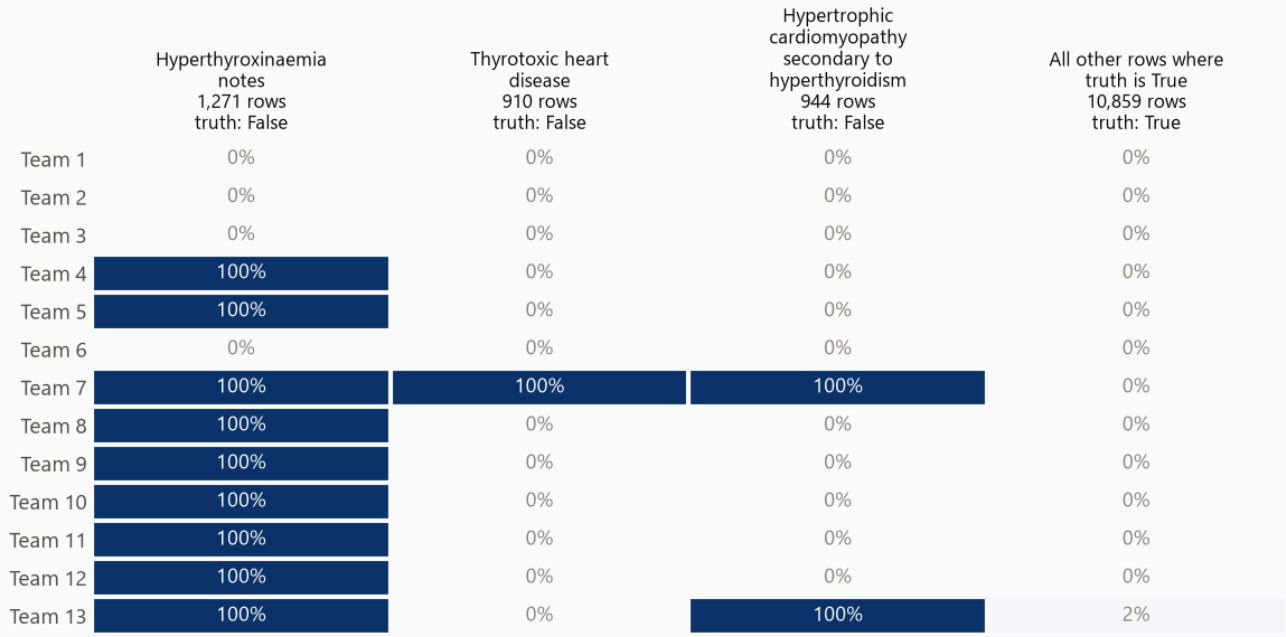
Finding 1 (Q4): "hyperthyroxinaemia" is not "hyperthyroidism"... or is it?

What and why. Q4 asks for hyperthyroidism "according to the clinical note". Two clinical note concepts are named as hyperthyroxinaemia: *Familial dysalbuminaemic hyperthyroxinaemia* and *Dysprealbuminaemic euthyroidal hyperthyroxinaemia*. The chart shows how each team answered on these rows, next to two other "looks like hyperthyroidism" groups.

```
1,271 rows carry a hyperthyroxinaemia note. Ground truth is True for 0 of them.
9 of 13 teams answered True on most of those rows: teams [4, 5, 7, 8, 9, 10, 11, 12, 13].
```

9 of 13 teams called hyperthyroxinaemia 'hyperthyroidism'

Share of rows where each team's answer differed from ground truth. Q4: hyperthyroidism according to the clinical note.



	Hyperthyroxinaemia notes	Thyrotoxic heart disease	Hypertrophic cardiomyopathy secondary to hyperthyroidism	All other rows where truth is True
Team 1	0.0	0.0	0.0	0.003
Team 2	0.0	0.0	0.0	0.003
Team 3	0.0	0.0	0.0	0.003
Team 4	1.0	0.0	0.0	0.003
Team 5	1.0	0.0	0.0	0.003
Team 6	0.0	0.0	0.0	0.000
Team 7	1.0	1.0	1.0	0.000
Team 8	1.0	0.0	0.0	0.003
Team 9	1.0	0.0	0.0	0.003
Team 10	1.0	0.0	0.0	0.003
Team 11	1.0	0.0	0.0	0.003
Team 12	1.0	0.0	0.0	0.003
Team 13	1.0	0.0	1.0	0.024

Result. 1,271 rows carry a hyperthyroxinaemia note. Ground truth is False for every one. Nine of 13 teams (4, 5 and 7 to 13) answered True on all of them. Teams 1, 2, 3 and 6 answered False. Team 7 went further and also answered True on the thyrotoxic heart disease rows, and Teams 7 and 13 did so on the hypertrophic cardiomyopathy rows.

Why this matters for a SNOMED CT audience. On Ontoserver (default edition at the time of checking), `34486009 |Hyperthyroidism|` subsumes `237547004 |Familial dysalbuminaemic hyperthyroxinaemia|`. A plain "descendants of hyperthyroidism" query therefore returns these concepts, while the competition's ground truth treats them as not hyperthyroidism. We do not pick a side. The ground truth defines the score, but this is a real tension between what the hierarchy says and how a clinician reads a euthyroid state. We checked that subsumption for the first concept only, in one edition, and did not check each of the six editions.

The organisers expected trouble of this kind. `Question_Rationale.md` says that "not every coded diagnosis is formally hyperthyroidism" and that a precise answer "needs careful research". It also names a

different problem on the same question, the 2022 move of *Thyrotoxicosis* out from under *Hyperthyroidism*. The question was rated 4 and was fifth hardest by outcome, so the rating held up.

Confidence. High on the counts. Medium on the interpretation, because we have not seen the teams' queries. The seven identical answer columns on Q4 (see "Answer twins") support the idea of one shared rule.

Finding 2 (Q15, also Q12): the same concept, a different answer in different editions

What and why. The data spans six SNOMED CT-AU editions, and each encounter is meant to be answered with the edition current on its date. If a concept moved in or out of the "pregnancy" grouping between editions, the right answer for the same note text changes with the year. We test that on Q15 (pregnant, fasting glucose of 5.1 or more). Restricting to rows with glucose of 5.1 or more removes the glucose condition, so any change in ground truth across years for one concept reflects the pregnancy definition.

We first count concepts whose ground truth flips between years, then compare how well teams track those flips.

```
7 clinical note concepts have ground truth that is True in some years and False in others (Q15, glucose >= 5.1, at least 3 rows per year).
They cover 228 rows, of which ground truth is True for 96.
Share of those rows each team answered correctly:
```

	Team 1	Team 2	Team 3	Team 4	Team 5	Team 6	Team 7	Team 8	Team 9	Team 10	Team 11	Team 12	Team 13
share correct on these rows	0.943	1.0	0.807	0.741	0.728	0.421	0.421	0.452	0.452	0.728	0.579	0.452	0.452

```
Rows with fasting glucose below 5.1: 198,955. Rows where any of the top 10 teams answered True: 0.
So the 5.1 threshold itself is applied consistently; the disagreement is about which notes count as pregnant.
```

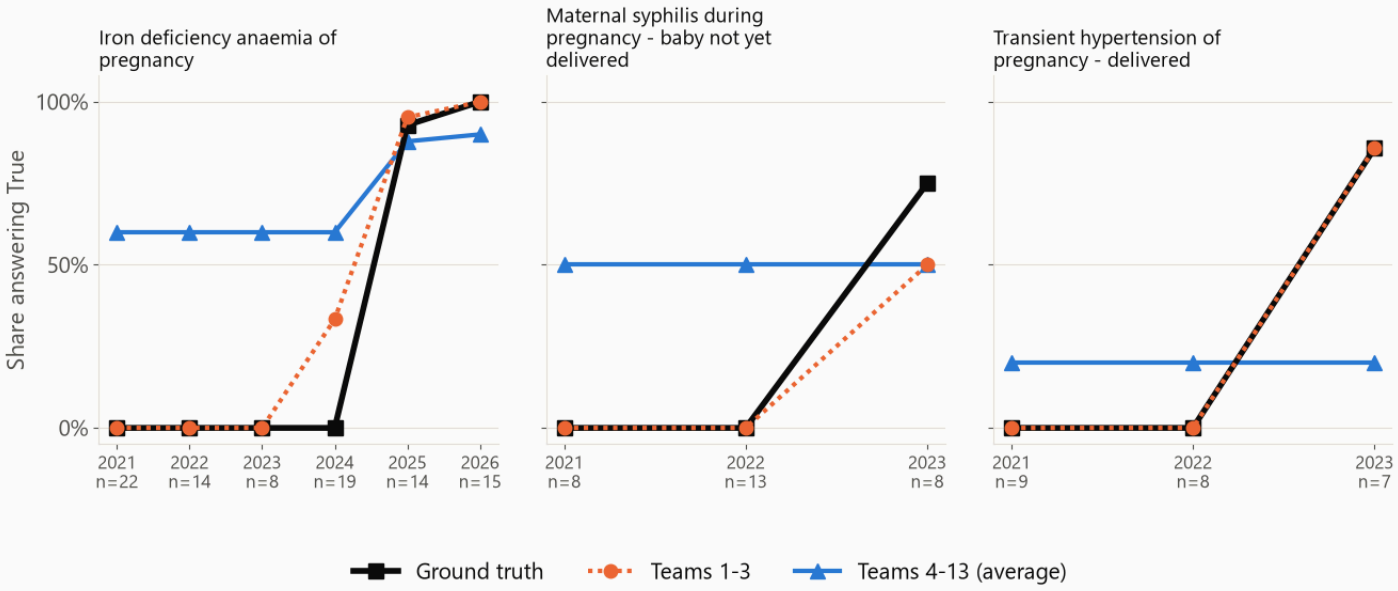
Result. Seven clinical note concepts have ground truth that is False in earlier years and True later (or the reverse), on 228 rows. Teams 1 to 3 answer 94%, 100% and 81% of those rows correctly. Teams 4, 5 and 10 manage about 73% to 74%, and the other seven teams sit between 42% and 58%, close to chance. The chart shows three examples: ground truth (black) steps up with the edition, Teams 1 to 3 (orange) mostly follow it, and the average of Teams 4 to 13 (blue) stays at one level across 2021 to 2024. The fasting glucose threshold is not the cause: no top 10 team answered True below 5.1.

Why this matters for a SNOMED CT audience. The question asked for each encounter to be answered with the edition current on its date. The same note text can sit inside or outside a "pregnant" grouping depending on the edition. The organisers' own comparison in `Question_Rationale.md` supports this: a pregnancy ECL returned between 775 and 1,602 concepts across the six editions, a 52% spread. Teams that used one fixed edition gain nothing from the extra years and lose accuracy on the older rows. Q12 (pregnant and teratogenic medicine) shows the same pattern, for example on *Acute fatty liver of pregnancy*, but the medicine condition confounds it, so we have not quantified it.

Confidence. Medium. It rests on seven concepts and 228 rows, with 7 to 22 rows per concept-year, though the three examples all behave the same way. We cannot see which edition each team queried. The flat blue lines suggest a single edition for most of them, but that is an inference. See also `../TerminologyDrift.md`.

Same concept, different answer by year

Q15 (pregnant, fasting glucose 5.1 or more), rows with glucose 5.1 or more. n = rows in that year.
Ground truth flips with the edition. Teams 1-3 largely follow it; Teams 4-13 barely change with the year.



Finding 3 (Q14): "digital" and "finger" in a name are not the same as the finger in the model

What and why. Q14 asks for an open wound on an upper limb, excluding the fingers and thumb. Some clinical notes contain the words *digital* or *finger* but, in ground truth, are counted as upper limb rows that are not finger or thumb rows. Other notes about thumb and finger amputation are not. The chart compares each team's answers across these groups.

Q14: each team tripped on a different finger

Share of rows where each team's answer differed from ground truth. Q14: open wound on an upper limb, excluding fingers and thumb.

	Open injury, digital vein or artery 1,036 rows truth: True	Open fracture, finger metacarpal 205 rows truth: True	Traumatic amputation of thumb and/or fingers 1,068 rows truth: False	All other rows where truth is True 21,359 rows truth: True
Team 1	0%	0%	0%	0%
Team 2	0%	0%	0%	0%
Team 3	0%	0%	0%	0%
Team 4	0%	0%	0%	0%
Team 5	0%	0%	0%	0%
Team 6	2%	6%	0%	0%
Team 7	100%	0%	0%	1%
Team 8	100%	100%	0%	1%
Team 9	0%	0%	100%	0%
Team 10	0%	0%	0%	0%
Team 11	0%	0%	100%	0%
Team 12	0%	100%	0%	1%
Team 13	51%	100%	0%	3%

	Open injury, digital vein or artery	Open fracture, finger metacarpal	Traumatic amputation of thumb and/or fingers	All other rows where truth is True
Team 1	0.000	0.000	0.0	0.000
Team 2	0.000	0.000	0.0	0.000
Team 3	0.000	0.000	0.0	0.000
Team 4	0.000	0.000	0.0	0.000
Team 5	0.000	0.000	0.0	0.000
Team 6	0.023	0.063	0.0	0.004
Team 7	1.000	0.000	0.0	0.008
Team 8	1.000	1.000	0.0	0.008
Team 9	0.000	0.000	1.0	0.000
Team 10	0.000	0.000	0.0	0.000
Team 11	0.000	0.000	1.0	0.000
Team 12	0.000	1.000	0.0	0.006
Team 13	0.506	1.000	0.0	0.030

Result. Teams 1 to 5 and 10 are perfect on every group here. The other teams each slipped differently:

- Teams 7 and 8 answered False on 1,036 *Open injury, digital vein or artery* rows, where ground truth is True. Team 13 did so on about half of them.
- Teams 8, 12 and 13 answered False on 205 *Open fracture, finger metacarpal* rows, where ground truth is True.
- Teams 9 and 11 answered True on 1,068 *Traumatic amputation of thumb and/or fingers* rows, where ground truth is False.

No failure is shared by more than three teams. This contrasts with Q4 and Q5, where seven or eight teams made one identical error.

Why this matters for a SNOMED CT audience. The words "digital" and "finger" in a concept name do not map cleanly onto the exclusion of "fingers and thumb". Reading by name goes wrong in both directions: ground truth keeps *finger metacarpal* fractures and digital vessel injuries in, and keeps thumb and finger amputations out. `Question_Rationale.md` describes the intended ECL as finding open wounds in the upper limb and then subtracting the digits, which is a statement about modelled anatomy and not about the words in a term. We have not checked how these concepts are modelled.

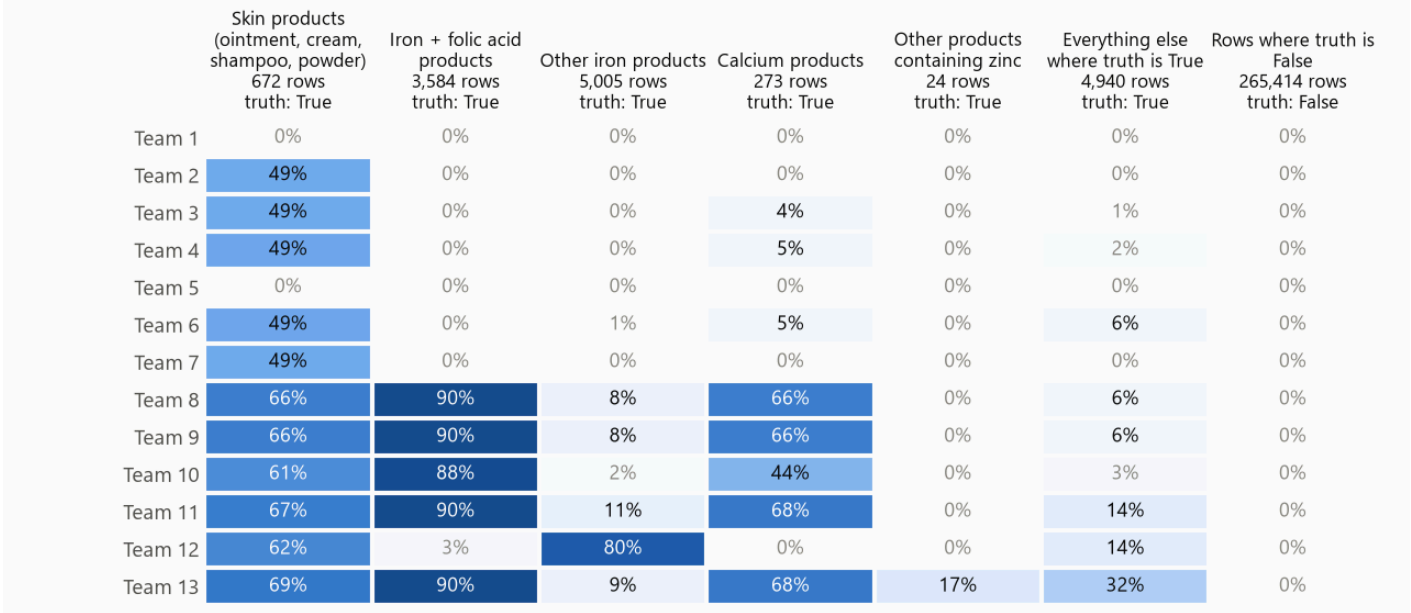
Confidence. Medium. The groups are found by name and are small (205 to 1,068 rows). The reasons teams slipped are hypotheses. The question was rated 3 and was tenth hardest by outcome.

Finding 4 (Q18): "mineral supplement" in the question, ingredients in the answer

What and why. Q18 asks for patients taking a mineral supplement, excluding sodium, potassium and chloride. Ground truth is True for many products that are not supplements in everyday terms (nappy-rash ointments containing zinc oxide, medicated shampoos, iron-containing phosphate binders). We group the ground-truth-True rows by the words in the medicine names to see which kinds of product each team captured. The groups are **approximate**: they come from keywords, not from the terminology.

Which mineral products did each team count?

Share of rows where each team's answer differed from ground truth. Q18: mineral supplement, excluding sodium, potassium or chloride. Groups are approximate (keywords in medicine names).



	Skin products (ointment, cream, shampoo, powder)	Iron + folic acid products	Other iron products	Calcium products	Other products containing zinc	Everything else where truth is True	Rows where truth is False
Team 1	0.000	0.000	0.000	0.000	0.000	0.000	0.001
Team 2	0.485	0.000	0.000	0.000	0.000	0.000	0.000
Team 3	0.488	0.000	0.004	0.044	0.000	0.006	0.000
Team 4	0.490	0.000	0.000	0.048	0.000	0.019	0.000
Team 5	0.000	0.000	0.000	0.000	0.000	0.000	0.002
Team 6	0.493	0.000	0.010	0.048	0.000	0.058	0.000
Team 7	0.485	0.000	0.004	0.000	0.000	0.000	0.001
Team 8	0.664	0.897	0.083	0.663	0.000	0.056	0.000
Team 9	0.664	0.897	0.083	0.663	0.000	0.056	0.000
Team 10	0.609	0.880	0.019	0.436	0.000	0.030	0.000
Team 11	0.670	0.897	0.109	0.678	0.000	0.141	0.000
Team 12	0.624	0.031	0.803	0.004	0.000	0.139	0.001
Team 13	0.690	0.898	0.093	0.681	0.167	0.322	0.000

Result. Ground truth is True on 14,498 rows, including 672 skin products (for example nappy-rash ointments with zinc oxide) and 3,584 iron-plus-folic-acid products.

- Teams 1 and 5 capture every group. Teams 2, 3, 4, 6 and 7 are right on iron and folic acid products but miss about half of the skin products (49%).
- Teams 8, 9, 10, 11 and 13 miss about 90% of the iron-plus-folic-acid products and about two thirds of the calcium products (44% for Team 10).
- Team 12 shows the reverse pattern: it captures the iron-plus-folic-acid products (3% missed) but misses 80% of other iron products.

Why this matters for a SNOMED CT audience. The question says "mineral supplement", but ground truth follows the ingredient. A reading by clinical intent drops zinc-oxide ointments and shampoos. A "mineral

only" reading drops combination products that also contain folic acid. The organisers expected this: `Question_Rationale.md` rates Q18 at 5, says the wording follows ATC group A12 but is "ambiguous and open to interpretation", and says variation was expected and accepted. It also notes that SNOMED CT has no "is modification of" link from an element to its salts and compounds, so a query on the element alone misses the compounds that are actually prescribed. We have not tested whether that explains the pattern here.

Confidence. High on the counts. Medium on the group boundaries, which come from keywords in medicine names. Low on the mechanism behind each team's pattern.

The shortlist at a glance

#	Finding	Chart	Expected difficulty, and observed rank (1 = hardest)	Why a SNOMED CT audience will care	Confidence
1	**Q4**: nine of 13 teams called euthyroid hyperthyroxinaemia "hyperthyroidism"	D1	4, observed 5th	A plain hierarchy query for hyperthyroidism includes concepts that the ground truth excludes. The hierarchy and the clinical meaning pull apart.	High on counts, medium on cause
2	**Q15 / Q12**: the same note concept has a different right answer in different editions	D2	Q15: 4, observed 2nd. Q12: 6, observed 1st	Edition-aware ECL matters. Only the top three teams track the changes.	Medium
3	**Q18**: "mineral supplement" in the question, ingredients in the answer	D4	5, observed 3rd	Wording and intent versus the ingredient model. Topical zinc oxide counts; iron plus folic acid products are the common miss.	Medium
4	**Q14**: each team tripped on a different finger	D3	3, observed 10th	Names with "digital" or "finger" do not map onto the model's anatomy. Errors are scattered, not shared.	Medium
5	**Difficulty ratings versus outcomes**	E	all 18 questions	The hard questions were the ones the organisers expected, but two questions rated 4 (Q10 and Q11) were easy and the middle of the scale is hard to predict.	Medium

Open question (not a finding). On Q5 (osteoarthritis in 2024), eight teams made one identical error: 340 rows in 2024, with gout- and chondrocalcinosis-type notes among the most common. We checked two of the concepts on Ontoserver in one edition, and neither is a descendant of *Osteoarthritis*. So the hierarchy as it stands today does not explain the shared error. `Question_Rationale.md` says the *Osteoarthritis* subtree shrank from 431 concepts in 2025 to 181 in 2026, and that concepts which stay active but move out of a subtree change answers more than retired concepts do. That makes edition handling a candidate explanation, but we have not tested it, so it stays out of the shortlist.

7. Caveats and checks

Caveats

- **Two submissions were excluded.** Teams 14 and 15 are treated as partial and left out after the leaderboard. That is our judgement from the answer patterns. It makes the other teams look more alike and the contested questions fewer. Set `EXCLUDE_BOTTOM = 0` to see the effect.
- **Placings are recomputed.** They use the competition's unweighted macro F1 over all rows. Confirm against the official leaderboard before presenting them as final. Teams 1 and 2 are effectively tied.
- **Difficulty ratings are expectations, not measurements.** They come from `Question_Rationale.md`: a six-point scale with many ties, 18 questions, and observed difficulty measured on 13 teams. The correlations in step 3 describe the pattern and are not a test of the organisers' judgement.
- **Sample size.** Thirteen teams is small. Clusters and "convergent" language are low to medium confidence.
- **We have not seen any team's method.** Every statement about why a team gave an answer is a hypothesis from the pattern of answers.
- **Keyword groups are approximate.** Findings 3 and 4 group patients by words in clinical note and medicine names.
- **Terminology checks were limited.** The Ontoserver checks used one edition and a handful of concepts. They do not cover all six editions.
- **Correctness** is defined only by the competition's solution file. Where a result points to a modelling ambiguity (Finding 1), we say so and do not choose a side.
- **Privacy.** No chart or table names a team. Only the last cell names the podium (Teams 1 to 3). Clear that cell's output before sharing the notebook more widely. The submissions folder carries real names in file names, so do not share that folder.

The cell below re-checks the main numbers against the files on disk and confirms that the saved charts and tables do not contain any participant name. The podium is revealed in the last cell only.

```
All checks passed. Slide images:
   outputs\A_team_by_question_heatmap.png
   outputs\B_contested_questions.png
   outputs\C_team_clustering.png
   outputs\E_expected_vs_observed_difficulty.png
   outputs\D1_Q4_hyperthyroxinaemia.png
   outputs\D2_Q15_edition_drift.png
   outputs\D3_Q14_digital_finger.png
   outputs\D4_Q18_ingredient_vs_intent.png
```

8. The podium

Every chart and table above uses anonymous labels. This last cell reveals who Teams 1 to 3 are and shows how close their scores were. Do not share the notebook outside the group that agreed to this without clearing the cell's output.

How close were they? Besides the gaps in macro F1 we ask how much of the gap could be noise. We resample the 279,912 patients with replacement 200 times, rescore the three teams on each resample, and count how often each team comes out ahead. This reflects sampling noise in this one dataset. It says nothing about how the teams would do on different questions.

Team	Name (from file name)	Macro F1 (all rows)	Public half	Private half	Rank on public half (of 13)	Rank on private half (of 13)
Team 1	Womp Ratter	0.9922	0.9918	0.9926	2	1
Team 2	Kenny-Bytes	0.9914	0.9919	0.9909	1	2
Team 3	Marcus Baw	0.9898	0.9906	0.9890	3	3

```
Gap in macro F1, Team 1 minus Team 2: all rows +0.0008, public -0.0001, private +0.0017
Gap in macro F1, Team 2 minus Team 3: all rows +0.0016
Gap in macro F1, Team 1 minus Team 3: all rows +0.0024
Cells where Team 1 and Team 2 gave different answers: 829 of 5,038,416 (0.0165%)
Cells where Team 1 and Team 3 gave different answers: 846 of 5,038,416 (0.0168%)
Questions where the top three differ in F1 by more than 0.01: Q5, Q12, Q13, Q15
Bootstrap over patients (200 resamples):
  Team 1 ahead of Team 2 in 92% of resamples; difference 2.5th to 97.5th percentile: -0.0004 to +0.0020
  Team 1 ahead of Team 3 in 100% of resamples
  Team 2 ahead of Team 3 in 98% of resamples
```

Question	Team 1	Team 2	Team 3	largest gap
Q12	0.9773	0.9913	0.9716	0.0197
Q15	0.9426	0.9538	0.8972	0.0566
Q18	0.9946	0.9881	0.9861	0.0085
Q16	0.9946	0.9915	1.0000	0.0085
Q13	0.9631	0.9503	0.9646	0.0142
Q17	1.0000	1.0000	1.0000	0.0000

Finding.

- **The podium** is Team 1 (Womp Ratter, macro F1 0.9922), Team 2 (Kenny-Bytes, 0.9914) and Team 3 (Marcus Baw, 0.9898), all within 0.0024 of each other.
- **Teams 1 and 2 are effectively tied.** Their order flips between the two halves: Team 2 leads the public half by 0.0001 and Team 1 leads the private half by 0.0017. The public gap is tiny and the private gap is larger, so the flip is lopsided and should not be over-read. The two teams gave different answers on only 829 of 5,038,416 cells (0.017%). In the bootstrap Team 1 is ahead in 92% of resamples, and the 95% range for the gap (-0.0004 to +0.0020) includes zero. The data do not separate them at the usual 95% level, though Team 1 has a slight edge.
- **Team 3 is a close but clearer third.** It is 0.0016 behind Team 2 and 0.0024 behind Team 1, and finishes behind them in 98% and 100% of resamples.
- **Where they differ.** The three differ in F1 by more than 0.01 on only four questions (Q5, Q12, Q13 and Q15), mostly the questions that were most contested overall.
- **Confidence: high** for the gaps, which are computed. The bootstrap reflects sampling noise in this one dataset. The answer key is fixed, so treat it as a guide to how fragile the ordering is, not as a significance test. The placings are recomputed, so check them against the official leaderboard.